

SHACHINDRAN A/L VEERANGAN

Software Systems | AI/NLP | Backend Engineering | Applied Machine Learning

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PROFILE

Information Technology undergraduate at Universiti Teknologi PETRONAS with hands-on AI/NLP R&D experience in Bangkok and independent software systems work. Built and tested document-processing pipelines across heterogeneous file formats, OCR, multimodal inference and asynchronous services, and am building GASOS for Malaysian LPG distribution operations. Interested in backend systems, document intelligence and applied AI that has to work reliably outside a demo.

EDUCATION

Universiti Teknologi PETRONAS (UTP) - Bachelor of Information Technology (Hons)

May 2024 - Expected Aug 2027 | Malaysia

Academic standing: Current CGPA 3.41/4.00 | Dean's List: 3 of 6 completed semesters

Bilkent University - Exchange Student, Computer Information Systems and Technologies (CTIS)

Sep 2025 - Jan 2026 | Ankara, Türkiye

Universiti Teknologi PETRONAS (UTP) - Foundation in Information Technology

2023 - 2024 | Malaysia

Academic standing: CGPA 3.78/4.00 | Dean's List in all three semesters | CFS Dean's Award

RESEARCH & TECHNICAL EXPERIENCE

Artificial Intelligence & Natural Language Processing Intern | Omniscien Technologies

May 2026 – August 2026 | Bangkok, Thailand

- Expanded document routing and metadata support from 28 to 100 file formats, independently validating 84 end-to-end across office, archive, image, geospatial, medical and specialist formats; the work was merged into the project's development repository.
- Benchmarked IBM Docling against Unstructured on complex PDFs with tables, formulas and multi-column layouts, then used the findings to build hybrid routing between faster and higher-fidelity extraction paths.
- Separated GPU-bound vision-language inference from CPU-bound extraction across isolated containerized workers, with routing between conventional OCR and local/remote multimodal analysis, structured-output validation and fallback handling.
- Expanded email ingestion and normalization to 11 validated formats and redesigned the pipeline around a canonical intermediate representation to remove duplicate extraction; the revised pipeline passed 97/97 automated tests.
- Resolved reliability issues across the pipeline, including a customer-reported Aspose.PDF watermark defect traced to a vector Form XObject, OCR-library dependency conflicts and repeated model-initialization overhead in isolated workers.

SELECTED TECHNICAL PROJECTS

GASOS - Independent Software Systems Project

2026 - Present | TypeScript, NestJS, React, PostgreSQL, Kysely, Docker, OpenAPI

- Designing and implementing the platform foundation for an operational system aimed at Malaysian LPG distribution businesses, using a modular monolith with a NestJS API, React application, PostgreSQL and a generated TypeScript API client.
- Implemented server-managed authentication, application user and tenant membership boundaries, roles and permissions, reviewed Kysely migrations, and an OpenAPI-driven browser client contract.
- Built an isolated integration-test workflow that creates a unique temporary PostgreSQL database per run, applies migrations and prevents tests from touching persistent development or pilot data.

Predictive Maintenance for Turbofan Engines - Independent Machine Learning Project [GitHub](#)

2026 | Python, scikit-learn, Pandas, NumPy | NASA C-MAPSS

Built an end-to-end Remaining Useful Life workflow with preprocessing, Random Forest regression, held-out engine evaluation, degradation analysis and reproducible visualizations.

Smart Energy Consumption Forecaster - Independent Time-Series Project

[GitHub](#)

2026 | Python, Prophet, Pandas, NumPy

Forecasted hourly American Electric Power demand data from 2004-2018 and analyzed daily, weekly and yearly seasonality to interpret recurring patterns and longer-term trends.

Additional projects

Wan and Whiskers: Unity/C#; 5-person team; contributed approximately 50% of the C# codebase, including multi-character control, destructible environments and endless mode.

Bank Loan Predictive Modeling: R, k-Nearest Neighbors; built and evaluated a classification workflow on 5,000 customer records to model loan-purchase propensity.

LEADERSHIP & COMMUNITY ENGAGEMENT

HoliRun UTP - Assistant Project Director

Jan 2026

- Served as second-in-command for a 250+ participant fundraising colour run with a RM29,390 approved budget, coordinating a 30-40 student committee, leading planning and monitoring deliverables while partly working remotely from Bilkent University.

UTP Sangeeth - Assistant Head, Event Management

2025 - 2026

Scope: Coordinated manpower, logistics, venues, schedules and event flow for cultural programmes; supported recruitment, CAVE promotions, Pongal programmes and team-building initiatives.

EduLink: Evolving Learning Experience - Media & Promotion Committee

Feb 2026

Impact: Managed promotion and event-media execution for a 20-member project at SMK Bercham, Ipoh, reaching 648 students and 71 teachers; produced campaign materials and live coverage.

Down to Earth - Sponsorship Committee (Mar 2024): Contributed to sponsorship outreach for an Ipoh community initiative; the committee secured RM4,230 in financial and in-kind contributions.

ADDITIONAL PROFESSIONAL EXPERIENCE

Horizon Advanced Tech Solutions - Sole Proprietor | Business Operations & Digital Administration

Dec 2023 - Present | Kuantan, Malaysia

Scope: Manage banking, invoicing, quotation review, bookkeeping coordination and tax documentation for a registered project-based technical services business; built and maintain its website and support digital operations.

Parkson - Part-Time Sales Promoter (Apr 2023; Mar 2026): Supported retail product promotion and day-to-day sales during short-term part-time engagements.

SCHOLARSHIPS, HONOURS & SELECTED AWARDS

YUTP Education Grant Recipient - Yayasan Universiti Teknologi PETRONAS (YUTP)

Sep 2023 - Present

Coverage: Merit- and need-based education grant covering full tuition fees for Foundation and undergraduate studies, with additional partial financial support for the Bilkent University exchange.

Selected awards: 1st Place - Tamil Public Speaking, Kuantan District Tamil Language Carnival (2022) | Five-Star Recognition - NILAM Reading Programme (2022) | Third Runner-Up - Manavar Mulakkam, Astro Vaanavil (2017, 100+ participants) | 5th Place - National Tamil Language Carnival (2017); participant again in 2018

TECHNICAL SKILLS & LANGUAGES

Programming: Python, TypeScript/JavaScript, Java, C#, R, SQL, HTML/CSS

Backend & data: NestJS, FastAPI, PostgreSQL, Kysely, Redis, SQLite, REST/OpenAPI, Celery

AI, ML & document intelligence: scikit-learn, Pandas, NumPy, Prophet, IBM Docling, Unstructured, Tesseract, PaddleOCR, multimodal/VLM inference, Azure AI Language

Infrastructure & additional: Docker, Kubernetes, Git, Linux, Pytest, Unity, MQTT, Wireshark, Autopsy, FTK Imager

Certifications: CS50's Introduction to Artificial Intelligence with Python (HarvardX, 2026) | Microsoft Applied Skills: Build a Natural Language Processing Solution with Azure AI Language (2026)

Languages: Tamil - Native | English - C1 | Bahasa Melayu - Fluent | Telugu - Basic, strong comprehension |

IELTS Academic 7.5 overall (L 7.5 | R 7.5 | W 6.5 | S 7.5), 2026